

Final

- TWO HOURS
- Cumulative

7 questions
1 worth 8 pts
others 7
evenly divided.

- 1st order ODEs

- Linear LINEAR
- sep
- exact

- systems $y' = Ay + b$

- y_c ~ real evcls
~ imaginary evcls
~ repeated evcl missing evcl

- y_p ~ b --- $\cos(3t)$ or $\sin(bt)$
~ b --- polynomials
~ b --- e^{at}

- IC in $y_c + y_p$

- higher order ODEs (lin, const-coeff)

- ~ char poly roots
 - real roots
 - complex roots
 - repeated roots

~ y_p

~ IC in $y = y_c + y_p$